

Numerical Heat Transfer Assignment

Report 1

Question 1 – Machine Error and Catastrophic Cancellation

Problem Description

Evaluate the effect of finite precision on the expressions:

$$D=(a+b)+c, E=a+(b+c) \quad D = (a+b)+c, \quad E = a+(b+c)$$

for two test cases, in both single and double precision, and compare with machine epsilon.

Method

- Implemented both expressions in Python (`float32` and `float64`).
- Computed machine epsilon.
- Compared DDD and EEE to quantify relative error.

Results

1. **Smaller test case** ($a=8888888$, $b=-8888887$, $c=0.3333341$) ($a=8888888$, $b=-8888887$, $c=0.3333341$):
 - Single precision: $D=1.3333341$, $E=1.0$ → Error $\approx 25\%$.
 - Double precision: $D=1.3333341$, $E \approx 1.3333340995$ → Error $\approx 3.7 \times 10^{-8}\%$.
2. **Larger test case** ($a=8888888888888888$, $b=-8888888888888887$, $c=0.3333341$) ($a=8888888888888888$, $b=-8888888888888887$, $c=0.3333341$):
 - Single precision: $D=0.3333341$, $E=0.0$ → Error = 100%.
 - Double precision: $D=1.3333341$, $E=1.375$ → Error $\approx 3.1\%$.

- **Machine epsilon:**

- float32: 1.19×10^{-7}
- float64: 2.22×10^{-16}

Discussion

- Errors are far larger than machine epsilon → caused by **catastrophic cancellation**, not single rounding.
- Demonstrates **loss of associativity** in floating-point arithmetic.
- The smaller test case shows double precision eliminates nearly all error, but the larger case shows that even double precision can suffer when the magnitudes of numbers differ drastically.

Question 2 – Numerical Differentiation of $\cos(x)$ at $x=0.2$

Problem Description

Approximate $\frac{d}{dx}\cos(x)$ at $x=0.2$ rad using forward and central differencing. Analyse truncation vs round-off errors.

Method

- Forward difference: first order, truncation error $O(\Delta x)$.
- Central difference: second order, truncation error $O(\Delta x^2)$.
- Errors computed against the analytical derivative $-\sin(0.2)$.
- Explored Δx from 10^{-20} to 0.1.

Results

- Forward error minimum: $\approx 6.23 \times 10^{-11}$ at $\Delta x \approx 1.05 \times 10^{-8}$.
- Central error minimum: $\approx 1.72 \times 10^{-12}$ at $\Delta x \approx 5.43 \times 10^{-6}$.
- Machine epsilon (float64): 2.22×10^{-16} .

Figures

- **Figure 1:** Total error vs Δx (linear).
- **Figure 2:** Truncation error vs Δx (log-log).
- **Figure 3:** Total error vs Δx (log-log).

Discussion

- Large $\Delta x \rightarrow$ truncation error dominates.
- Very small $\Delta x \rightarrow$ round-off dominates due to subtracting nearly equal numbers.
- Errors are minimised when truncation and round-off are balanced.
- Central differencing is more accurate overall, consistent with its second-order truncation error.

Question 3 – Steady-State Heat Conduction in a Plate

Problem Description

Solve 2D Laplace's equation on a rectangular steel plate with fixed boundary temperatures. Compare results for different grid resolutions and stencils.

Method

- Implemented finite-difference discretisation:
 - **5-point (second order).**
 - **9-point (fourth order).**

Building matrix and RHS for $L=8$ cm, $M=4$ cm, $\Delta x=2$ cm, $\Delta y=1$ cm...

Coefficient matrix A (5-point scheme):

`[[ -2.5 0.25 0. 1. 0. 0. 0. 0. 0. ]`

`[ 0.25 -2.5 0.25 0. 1. 0. 0. 0. 0. ]`

`[ 0. 0.25 -2.5 0. 0. 1. 0. 0. 0. ]`

`[ 1. 0. 0. -2.5 0.25 0. 1. 0. 0. ]`

`[ 0. 1. 0. 0.25 -2.5 0.25 0. 1. 0. ]`

`[ 0. 0. 1. 0. 0.25 -2.5 0. 0. 1. ]`

```
[ 0.  0.  0.  1.  0.  0. -2.5 0.25 0. ]
[ 0.  0.  0.  0.  1.  0.  0.25 -2.5 0.25]
[ 0.  0.  0.  0.  0.  1.  0.  0.25 -2.5 ]]
```

Right-hand side vector b:

```
[-600. -550. -650. -50.  0. -100. -350. -300. -400.]
```

Results

- Small plate: 9×9 system matrix and RHS built.
- Large plate: temperatures at (6,6), (10,6), (16,6) cm summarised below:

dx (cm)	dy (cm)	stencil	T(6,6) K	T(10,6) K	T(16,6) K	max err (K)
		I				
2.00	2.00	5pt	358.963	386.396	393.527	0.721
2.00	2.00	9pt	358.963	386.396	393.527	0.721
1.00	1.00	5pt	359.381	386.930	393.475	0.187
1.00	1.00	9pt	359.484	387.062	393.482	0.055
2.00	1.00	5pt	358.722	386.504	393.538	0.808
2.00	1.00	9pt	359.047	386.840	393.619	0.483
1.00	0.50	5pt	359.327	386.963	393.471	0.203
1.00	0.50	9pt	359.493	387.098	393.472	0.038

Figure 4 – Temperature contour (9pt, dx=1 cm, dy=0.5 cm).

Discussion

- Accuracy improves with smaller Δx and higher-order schemes.
- **Most accurate:** 9-point with $\Delta x=1$ cm, $\Delta y=0.5$ cm (error ≈ 0.038 K).
- **Practical conclusion (per professor):** The **3-point (5-point stencil) scheme with $\Delta x=\Delta y=1$ cm** already gives very good accuracy (error < 0.2 K), with much lower computational cost.

- Using $\Delta x = 2$ cm is acceptable for quick estimates, but accuracy decreases.
- Higher-order schemes are useful for verification but may not always be worth the extra complexity in engineering practice.

Summary of AI Interaction

I used ChatGPT as a coding assistant. It helped generate and run Python scripts to:

- Demonstrate floating-point non-associativity.
- Compute derivative errors and plot error curves.
- Build coefficient matrices and solve 2D conduction problems.

The interpretation, numerical comparisons, and final discussion were prepared by me, informed by both AI results and the professor's lecture notes.